NEWTON SCHOOL

PRE-CALCULUS 쨋 WORKSHEET

Polar Curves Gallery: Identify & Sketch

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Standard: *CCSS.MATH.CONTENT.HSF.TF ??Sketch polar curves: circles, cardioids, lima첼ons, roses, special edges.*

Common polar curves at a glance:

- **Circle** ??$r = a,\ \ r = a\sin\theta,\ \ r = a\cos\theta.$
- **Spiral (Archimedean)** ??$r = a\theta.$
- **Lima첼on** ??$r = a \pm b\sin\theta$ or $r = a \pm b\cos\theta$ $(a, b > 0)$. 4 sub-types based on a/b.
- **Rose** ??$r = a\sin n\theta$ or $r = a\cos n\theta$. n odd $\Rightarrow$ n petals. n even $\Rightarrow$ $2n$ petals.
- **Edge cases** ??$r^2 =$ trig (lemniscate-ish), $r = \sec\theta$ (line), $r = a/(1 \pm e\sin\theta)$ (conic).

Part A ??Identify the Curve Family

Directions: *Classify each polar equation by family and sub-type. No sketching yet ??pattern-matching only.*

NOTE 쨋 CLASSIFICATION RULES

Lima첼on $r = a \pm b \cdot \mathrm{trig}(\theta)$: compare a and b.

- $a < b \Rightarrow$ *inner loop.*
- $a = b \Rightarrow$ *cardioid* (heart shape, passes through pole).
- $a > b \Rightarrow$ *dimpled lima첼on* (small indentation).
- $a \geq 2b \Rightarrow$ *convex lima첼on* (smooth oval).

Rose $r = a\,\mathrm{trig}(n\theta)$: n odd $\Rightarrow$ n petals; n even $\Rightarrow$ $2n$ petals. Petal length $|a|$.

Worked Example ??Classify $r = 2 + 2\sin\theta$

Match pattern. Form is $r = a + b\sin\theta$ with $a = 2,\ \ b = 2$. Family = **lima첼on.**

Sub-type. $a = b \Rightarrow$ **cardioid.**

Orientation. $+\sin \Rightarrow$ max at $\theta = \pi/2$ (top) $\Rightarrow$ cardioid points **upward.**

A1. Classify each equation by family + sub-type (where applicable):

a. $r = 3 + 5\cos\theta$ — family: __________ sub-type: __________

b. $r = 4 - 2\sin\theta$ — family: __________ sub-type: __________

c. $r = 2\cos 3\theta$ — family: __________ # of petals: ____

d. $r = 5\sin 4\theta$ — family: __________ # of petals: ____

A2. Match each lima챌on equation to its orientation (up / down / left / right). Explain in one sentence how the *sign* and the *trig choice* (sin vs cos) determine orientation.

a. $r = 1 + \cos\theta$ — ____________

b. $r = 1 - \cos\theta$ — ____________

c. $r = 1 + \sin\theta$ — ____________

d. $r = 1 - \sin\theta$ — ____________

Rule (one sentence):

A3. Without sketching, predict the skeleton of $r = 2\cos 5\theta$: how many petals, and along which axis does the first petal point? Explain in 1 sentence.

Part B ??Sketch Circles & Cardioids

Directions: *Build a small table of (θ, r) values, then sketch on the polar grid. Connect smoothly.*

NOTE 쨋 ANATOMY

Circle $r = a\cos\theta$: passes through pole, diameter $|a|$, sits on the x-axis (sign of a chooses left/right). $r = a\sin\theta$ similar, on the y-axis (up/down).

Cardioid $r = a(1 \pm \text{trig}\,\theta)$: heart shape. Max distance $2a$ on one side, passes through the pole on the opposite side.

Worked Example ??Sketch $r = 1 + \cos\theta$ (cardioid pointing right)

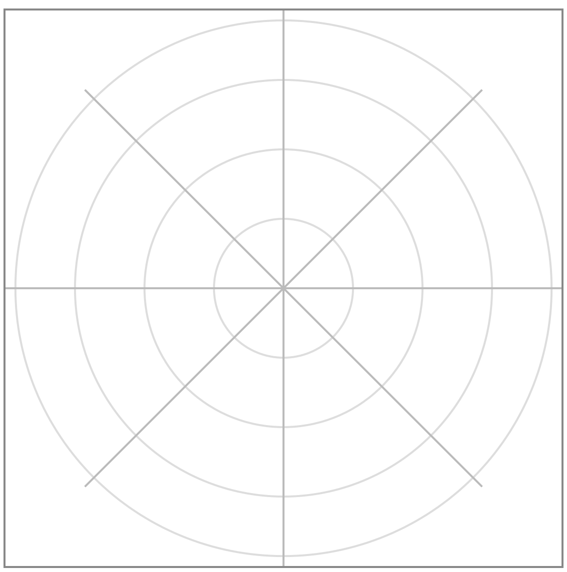
Table.

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π	$4\pi/3$	$3\pi/2$	$5\pi/3$
r	2	1.5	1	0.5	0	0.5	1	1.5

Shape. Max at $\theta = 0$ ($r = 2$ on positive x-axis). Passes through pole at $\theta = \pi$. Symmetric about x-axis. Heart points **right**.

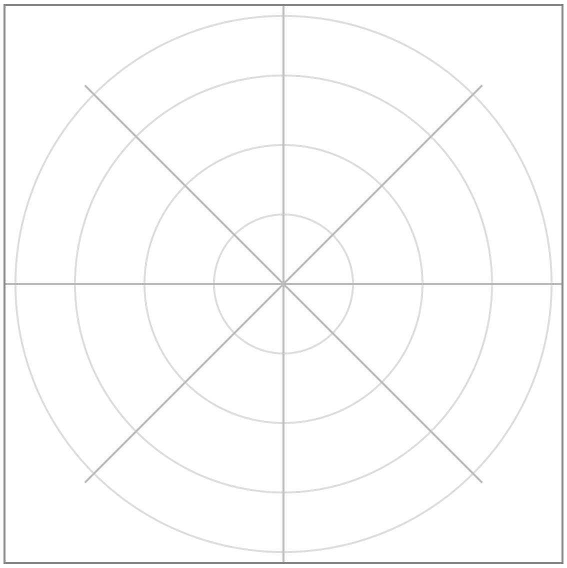
B1. Sketch $r = 4\sin\theta$. Identify diameter and axis.

Polar grid (each ring = 1):



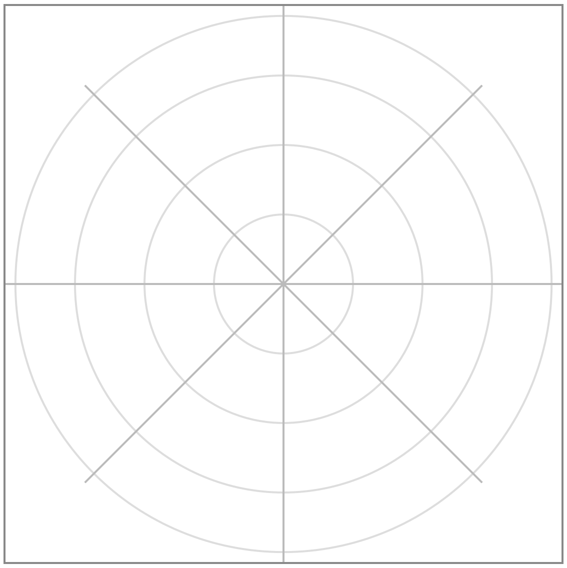
B2. Sketch $r = 6\cos\theta$. Identify diameter and which side of the y-axis.

Polar grid (each ring = 2):



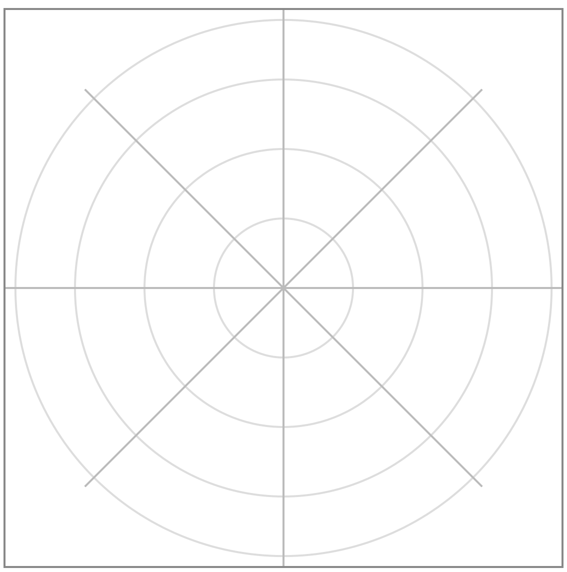
B3. Sketch $r = 3(1 - \sin\theta)$ (cardioid). Which direction does it point?

Polar grid (each ring = 1.5):



B4. Sketch $r = 2 + 2\cos\theta$ (cardioid pointing right). Compare to B3.

Polar grid (each ring = 1):



Part C ??Sketch Lima첼ons

Directions: *Compare a and b first to predict the sub-type. Then check key angles $\theta = 0, \pi/2, \pi, 3\pi/2$ for the max, min, and pole-crossings.*

NOTE 쨌 SUB-TYPES

Inner loop ($a < b$) — graph dips through the pole on the side opposite the main lobe and traces a small loop there.
Dimpled ($a > b$ but $a < 2b$) — small indentation, no loop.
Convex ($a \geq 2b$) — smooth oval.
Pole-crossings: solve $a \pm b\,\text{trig}(\theta) = 0$ to find the angles where $r = 0$??these are inner-loop entry/exit angles.

Worked Example ??Sketch $r = 1 + 2\sin\theta$

Classify. $a = 1,\ b = 2$, so $a < b \Rightarrow$ **inner loop**. $+\sin \Rightarrow$ main lobe points up.

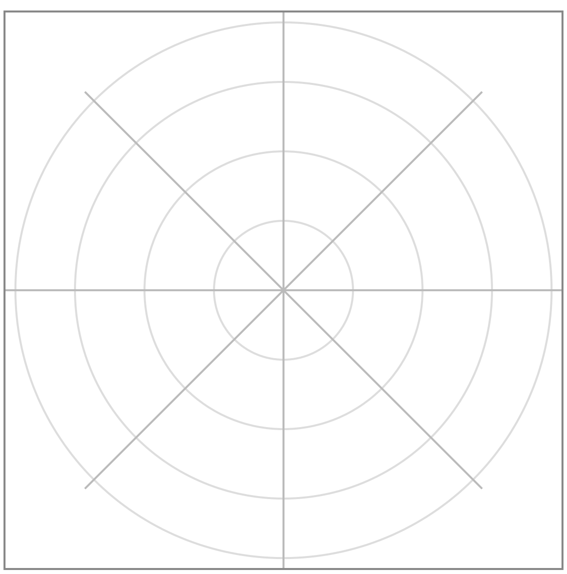
Key values. $\theta = 0$: $r = 1$. $\theta = \pi/2$: $r = 3$ (max, top). $\theta = \pi$: $r = 1$. $\theta = 3\pi/2$: $r = -1$ (negative ?? main bottom point flips, traces small loop).

Pole crossings. $1 + 2\sin\theta = 0 \Rightarrow \sin\theta = -1/2 \Rightarrow \theta = 7\pi/6,\ 11\pi/6$. Inner loop appears between these angles, on the bottom.

Shape. Big upward lobe $(max 3 at \theta = \pi/2)$ + small downward inner loop (max 1 below pole).

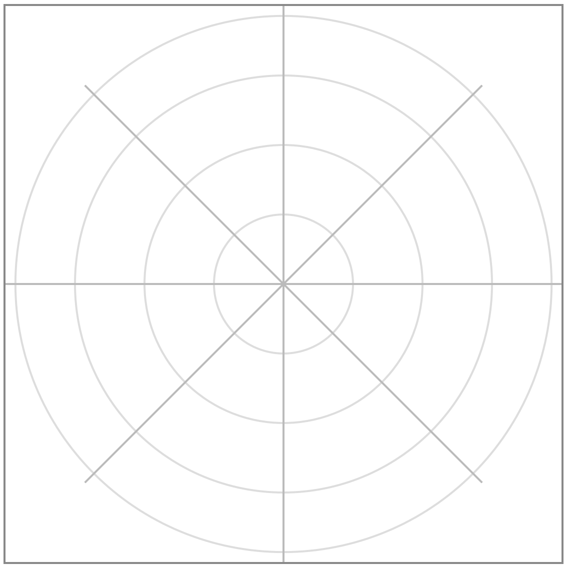
C1. Sketch $r = 2 - \cos\theta$. (Dimpled, $a > b$, oriented left because of $-\cos$.)

Polar grid (each ring = 1):



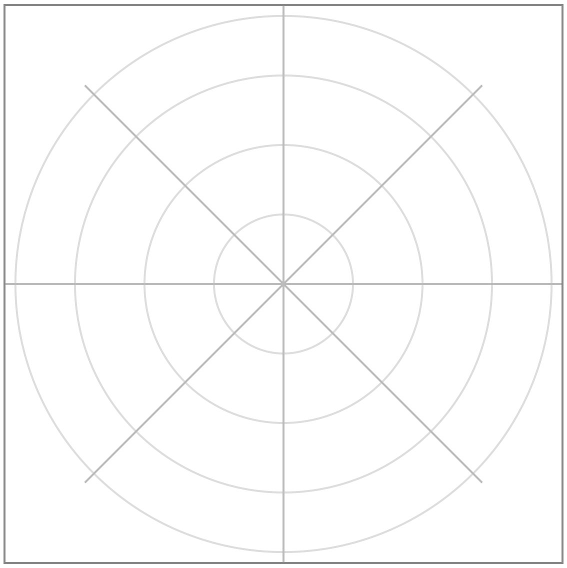
C2. Sketch $r = 1 - 2\cos\theta$. (Inner loop, $a < b$. Solve for pole crossings.)

Polar grid (each ring = 1):



C3. Sketch $r = 4 + 2\sin\theta$. (Convex, $a \geq 2b$. Smooth oval.)

Polar grid (each ring = 1.5):



Part D ??Sketch Roses

Directions: *Count petals first (odd $n \Rightarrow n$, even $n \Rightarrow 2n$). Then locate the first petal (cos $\Rightarrow$ along $\theta = 0$; sin $\Rightarrow$ along $\theta = \pi/(2n)$).*

NOTE 짹 ROSE ANATOMY

$r = a\cos n\theta$: first petal along positive x-axis, length $|a|$.
$r = a\sin n\theta$: first petal along bisector $\theta = \pi/(2n)$.
Petals are evenly spaced around the pole.

Worked Example ??Sketch $r = 3\sin 2\theta$

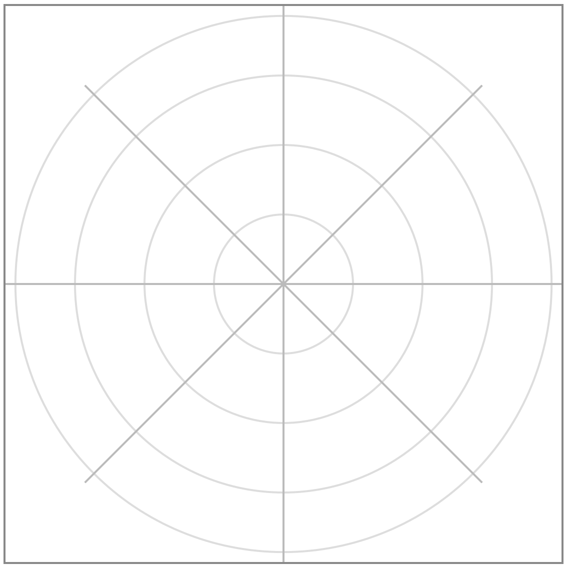
Petal count. $n = 2$ (even) $\Rightarrow 2n = 4$ petals. Length $= 3$.

First petal. $\sin 2\theta$ is max when $2\theta = \pi/2$, i.e. $\theta = \pi/4$. First petal points into Q-I along the 45째 bisector.

All petals. Petals at $\theta = \pi/4,\ 3\pi/4,\ 5\pi/4,\ 7\pi/4$ (one per quadrant, each along the quadrant bisector).

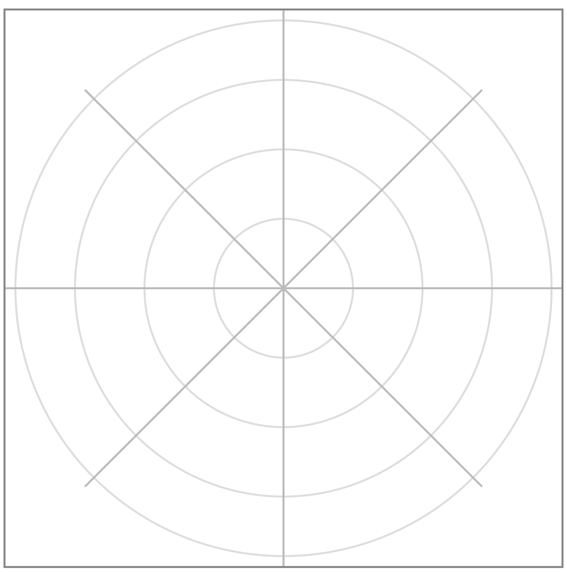
D1. Sketch $r = 2\cos 3\theta$. (3 petals.)

Polar grid (each ring = 1):



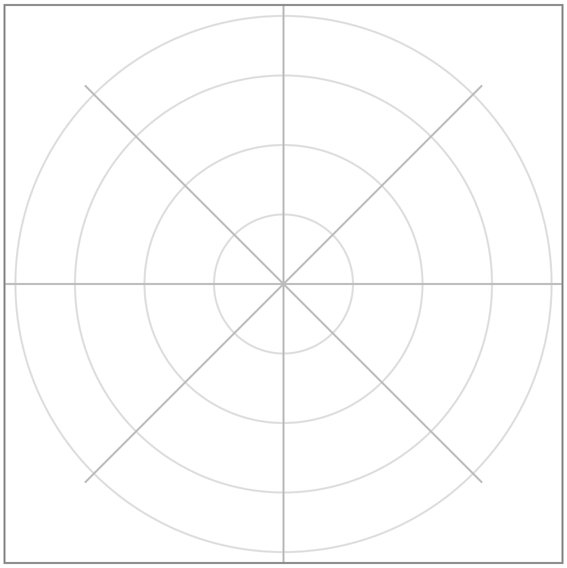
D2. Sketch $r = 4\sin 4\theta$. (8 petals.)

Polar grid (each ring = 1):



D3. Sketch $r = \sin 5\theta$. (5 petals.)

Polar grid (each ring = 0.5):



Part E ??Edge Cases

Directions: *These are non-standard. Simplify the equation algebraically first, then sketch.*

NOTE 짼 TRICKY FORMS

$r^2 = \text{trig}(\theta)$ forms restrict the domain to where the RHS ≥ 0 (since $r^2 \geq 0$). Result: *lemniscate-style* loops in only certain quadrants.
$r = \sec\theta$ or $r = \csc\theta$ collapses to a vertical/horizontal line in rectangular form.
$r = a/(1 \pm e\,\text{trig}\,\theta)$ is the polar form of a conic section (ellipse / parabola / hyperbola depending on e).

Worked Example ??Sketch $r^2 = \tan\theta$

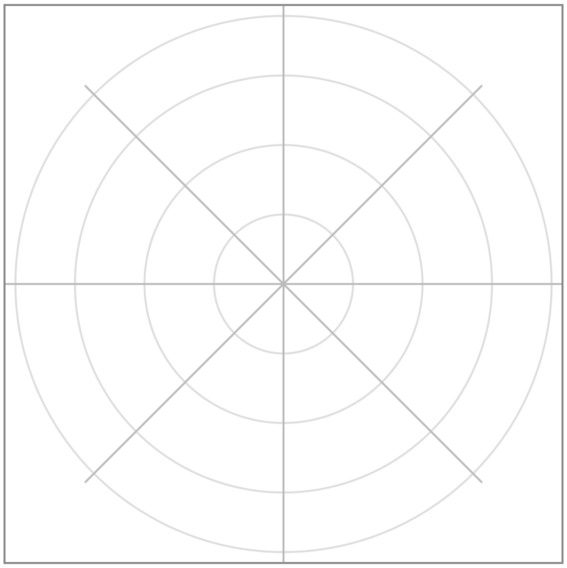
Domain. $r^2 \geq 0$ requires $\tan\theta \geq 0$, i.e. $\theta \in [0, \pi/2) \cup [\pi,\ 3\pi/2)$ (Q-I and Q-III).

Behavior. At $\theta = 0$, $r = 0$ (pole). As $\theta \to \pi/2^-$, $\tan\theta \to \infty$, so $r \to \infty$??curve shoots outward asymptotically. Symmetric repetition in Q-III.

Shape. Two loops/branches emerging from the pole, one in Q-I and one in Q-III, both unbounded along the y-axis direction.

E1. Sketch $r = \sec\theta$. *Hint:* rewrite as $r\cos\theta = 1$, then convert.

Polar grid (each ring = 1):



E2. Sketch $r = \dfrac{4}{1 + 2\sin\theta}$. *Hint:* identify e (the eccentricity) and find where the denominator is zero (asymptotic direction).

Polar grid (each ring = 2):